

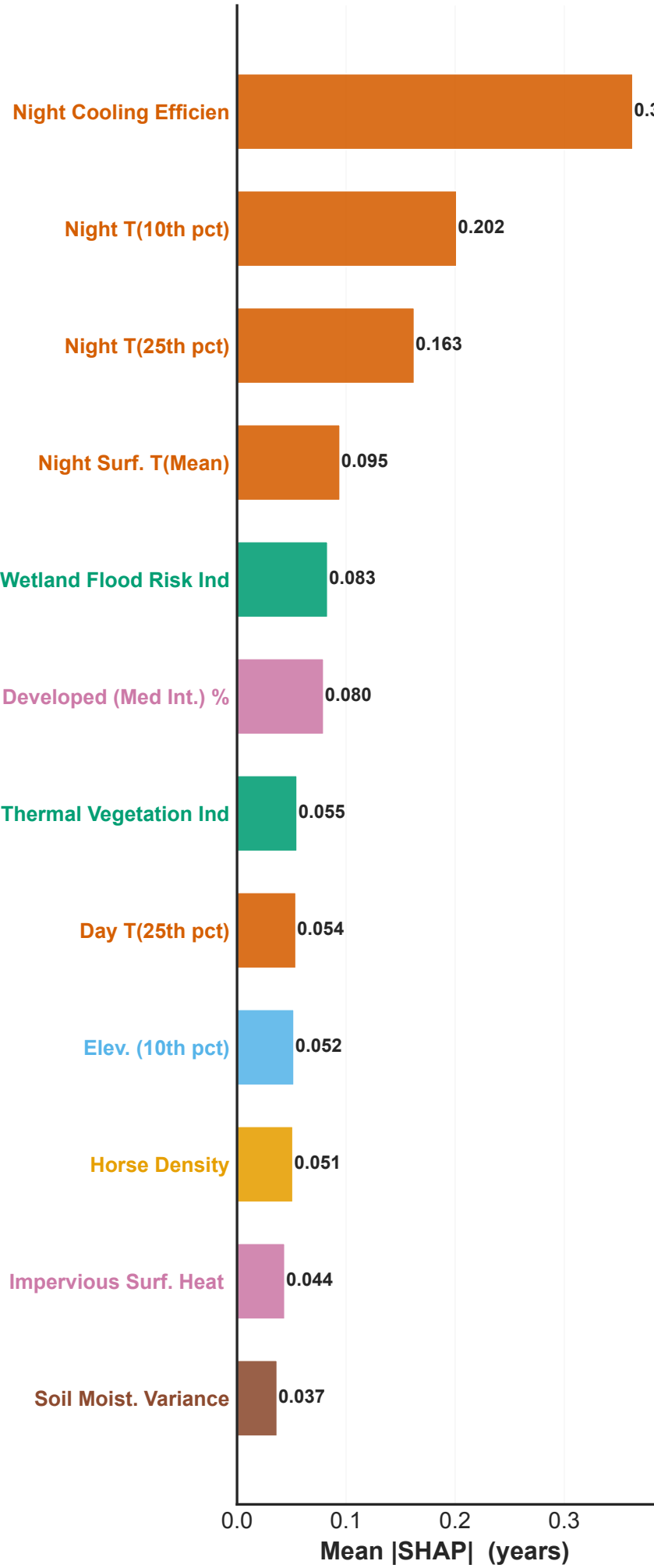
Cross-Modal SHAP Interaction Structure

Correlation of feature importance patterns reveals synergistic & antagonistic modality pairs

(D1) |SHAP| correlation matrix (tick colour = modality)



(D2) Individual mean |SHAP|



Nighttime LST features form a strongly correlated cluster ($r > 0.9$)
— indicative of a unified thermal cooling mechanism.

Modality

Temperature Vegetation Topography
Development Livestock Soil